

Introduction

The constantly evolving financial landscape requires financial institutions and stakeholders to adopt a data-driven approach to maintain continued success. Companies who lack the ability to translate raw, senseless into actionable insights would find themselves significant behind the competition and even run out of business. One dynamic institution focused on exceptional service and maintaining market dominance, Aurora Bank, recognizes the state of the financial industry and the need to stay on top by tapping into the inherent value within its data resources. Out of this data-driven strategy comes this project, which delves into the patterns of Aurora Bank's customer behaviour, financial health, and financial activity. Through SQL, we aim to analyze the customer demographics, financial health, transaction behaviours, and risks.

Objectives

- ☐ Develop a structured relational database using Microsoft SQL Server to store and manage banking data.
- ☐ Perform in-depth analysis to understand customer profiles, spending habits, and potential risk factors.
- ☐ Identify fraudulent patterns and transaction errors.
- ☐ Provide actionable recommendations for improving financial risk management and customer engagement.

2. Data Architecture

The database is organized using a relational model made from SQL Server, with four main schemas:

2.1 Cards Data Schema

Column Name	Data Type	Description
id	SMALLINT PK	Unique identifier for each credit card.
client_id	SMALLINT (FK), NOT NULL	Links the card to its owner in the users data table.
card_brand	NVARCHAR(50), NOT NULL	Brand/issuer of the credit card (e.g., Visa, Mastercard).
card_type	NVARCHAR(50), NOT NULL	Type/category of the card (e.g., debit, credit, debit(prepaid)).
card_number	BIGINT, NOT NULL	Unique number on the credit card.
expires	NVARCHAR(50), NOT NULL	Expiry date of the credit card.
cvv	SMALLINT, NOT NULL	Security code of the card.
has_chip	BIT, NOT NULL	Indicates if the card has an EMV chip.

num_cards_issued	TINYINT	Total number of cards assigned for the same account.
credit_limit	MONEY, NOT NULL	Credit limit assigned to the card.
acct_open_date	NVARCHAR(50), NOT NULL	Date the credit card was opened.
year_pin_last_changed	SMALLINT, NOT NULL	Year the card's PIN was last changed.

2.2 Users Data Schema

Column Name	Data Type	Description
id	SMALLINT (PK)	Unique identifier for each user.
current_age	TINYINT, NOT NULL	Current age of the user
retirement_age	TINYINT, NOT NULL	Declared retirement age of the user.
birth_year	SMALLINT, NOT NULL	Year of birth for the user
birth_month	TINYINT, NOT NULL	Month of birth for the user
gender	NVARCHAR (50), NOT NULL	Gender of the user (male or female)
address	NVARCHAR (50), NOT NULL	Residential address of the user
latitude	NVARCHAR (50), NOT NULL	Latitude of the user's residence
longitude	FLOAT, NOT NULL	Longitude of the user's residence
per capita income	MONEY, NOT NULL	Per capita income of the user.
yearly_income	MONEY, NOT NULL	Total yearly income of the user.
credit_score	SMALLINT, NOT NULL	Credit score rating of the user
num_credit_cards	TINYINT, NOT NULL	Number of credit cards owned by the user.

2.3 Merchant Category Codes (MCC) Schema

Column Name	Data Type	Description
mcc_id	SMALLINT (PK), NOT NULL	Unique identifier for each Merchant Category Code.
description	NVARCHAR(100), NOT NULL	Description of the merchant category (e.g., grocery, travel, entertainment)

2.4 Transactions Data Schema

Column Name	Data Type	Description
id	INT (PK), NOT NULL	Unique identifier for each transaction.
date	Datetime 2(7), NOT NULL	Date of the transaction
client_id	SMALLINT (FK), NOT NULL	Links the transaction to the user in the users data table
card_id	SMALLINT (FK), NOT NULL	Links the transaction to the respective card in the cards data table.
amount	FLOAT, NOT NULL	Transaction amount in the respective currency.
use_chip	NVARCHAR(50), NOT NULL	Indicates if the chip was used during the transaction.
merchant_id	INT, NOT NULL	Unique identifier of the merchant.
merchant_city	NVARCHAR(50), NOT NULL	City where the merchant is located.
merchant_state	NVARCHAR(50), NULL	State where the merchant is located
zip	INT, NULL	Zip code of the location
mcc	SMALLINT (FK), NOT NULL	Merchant category code representing the type of merchant or transaction.
errors	NVARCHAR(50), NULL	Any errors encountered during the transaction (if any)

The project is divided into two areas of focus, which were thoroughly covered:

- **Customer Profiling & Segmentation:** This section looks into the demographic and financial characteristics of Aurora Bank's clientele. With this information, Aurora can know how to personalize their services, enhance credit risk assessment methodologies, and optimize financial planning services: Key areas tackled include:
 - o **Demographic Distribution:** Understanding the gender and age distribution of the customer base to effectively tailor the bank's financial products.
 - o **Geographic Insights:** Identifying critical locations of customer concentration to optimize resource allocation and outreach efforts.
 - o **Financial Health Assessment:** Evaluating yearly income, per capita income, and debt-to-income ratios to gain a deeper understanding of customer financial stability.
 - o **Credit Risk Analysis:** Understanding credit risk levels across different customer segments to inform lending practices and risk management strategies.

- **Transaction and Risk Analysis:** This section looks into customer spending behaviours while simultaneously identifying potential financial risks and indicators of fraudulent activity. Key areas tackled include:
 - **Spending Patterns & Merchant Categories:** Analyzing transaction volume across diverse categories (e.g., retail, travel, dining) to understand customer spending habits and preferences.
 - **Transaction Anomalies & Fraud Indicators:** Monitoring failed transactions, high-value purchases, and unusual spending behaviours to proactively identify and mitigate potential risks.
 - **Debit and Credit Risk Management:** Assessing total debt distribution, credit utilization patterns, and identifying customers who may be at risk of default to inform proactive intervention strategies and portfolio management.

By incorporating insights gained from this project, Aurora Bank can continue to excel by refining its financial strategies and services, optimize risk management methodologies, improve customer engagement, and maintain brand loyalty. The following sections will provide detailed breakdowns, supported by SQL-driven analysis, ensuring that the bank can make informed, data-backed decisions and thrive in the financial industry.

Timeframe and Transaction Volume

Spanning a two-year period, the dataset encompasses a total of 157,224 recorded transactions. This extensive timeframe provides a solid foundation for a comprehensive analysis of customer activity over time. It allows for a deeper exploration of spending patterns, financial behaviour, and segmentation based on key factors such as age, income, and credit usage. Understanding this overall transaction volume and the two-year timeframe is crucial as it provides the necessary context for interpreting subsequent insights, revealing the frequency of customer engagement with financial services and the evolution of their behaviours over time.

Implications:

- While a two-year period provides a broad view, it's important to consider potential seasonal variations or economic fluctuations that may have influenced transaction patterns within that timeframe. Without more information, it is difficult to see trends.
- **Operational Load:** The sheer volume of transactions implies a significant operational load on the bank's systems, requiring robust infrastructure and efficient processing capabilities.

Transaction Value

Over the two-year period, the total value of transactions reaches approximately **\$6.87 million**. This figure reflects the overall scale of financial activity within the dataset and provides valuable insight into the collective spending power of our collective base. When considered alongside analyses such as average spend per customer segment and transaction, this total transaction becomes instrumental in understanding purchasing power, liquidity, and engagement levels across various demographic groups.

Implications:

- **Uneven Distribution:** The total value does not reveal how spending is distributed across customer segments, potentially masking significant disparities.
- **Missed Growth Opportunities:** Without trend analysis, the data does not indicate if transaction value is increasing, decreasing, or stagnant, hindering strategic growth planning.
- **Limited Customer Insight:** Relying solely on total value overlooks the diverse purchasing behaviours and liquidity needs of different customer demographics.

The questions raised by the implications were answered in the rest of the project.

.Distribution of Transaction Types

Chip-based transactions dominate the total volume, with **112,114 recorded instances**, significantly exceeding swipe transactions (**27,327**) and online transactions (**17,783**). This indicates a strong preference for or requirement to use chip-based authentication for the majority of transactions, likely due to the enhanced security offered by EMV chip technology. Swipe transactions, which are less secure, are notably lower in frequency. Online transactions are the least frequent, potentially reflecting customer behaviour, security concerns, or limitations in online transactions capabilities.

Implications:

- **Low Adoption of Other Transaction Methods:** The dominance of chip transactions (71% of all transactions) raises questions about the accessibility and adoption of other transactions methods, particularly online transactions.
- **An Opportunity for Improvement:** The low volume of online transactions might indicate a need to improve the online transaction experience, address security concerns, or expand online transaction capabilities. And given that many customers are young, it becomes essential.
- **Transaction Types Might Be Influenced by Proximity:** The large amount of chip transactions could be a sign of the physical locations of the customers and the type of businesses they frequent.

Total Transaction Value by Type

Chip transactions not only represent the highest volume but also account for the largest share of total transaction value, amounting to approximately **\$4.64 million**. This reinforces the earlier observation that chip-based authentication is the dominant method, likely driven by security preferences or regulatory requirements. Swipe transactions follow with a total value of around **\$18 million**. Notably, despite being the least frequent, online transactions contribute a comparable amount of **\$1.06 million**, suggesting that online purchases typically involve higher average values compared to swipe transactions.

Implications:

- ❑ **Opportunity for Expansion:** The comparable total value of online transactions despite their lower frequency suggests a potential for growth in this channel if barriers to adoption can be overcome.
- ❑ **Customers Prefer Physical Transactions:** The fact that chip transactions have the highest total value shows that the main way customers are spending money is in person.

Average Transaction Value by Type

Online transactions exhibit the highest average value, approximately **\$59.35**, suggesting a tendency for customers to make larger purchases when transacting online. Swipe transactions follow with an average value of **\$43.06**. Interestingly, while chip transactions are the most frequent and contribute the highest total value, they have the lowest average value at **\$41.41**. This indicates that chip transactions are generally characterized by smaller but more frequent purchases, whereas online transactions, though less common, involve higher-value items.

Implications:

- ❑ **Everyday Purchases the Norm for Customers:** The lower average value of chip transactions could indicate a high volume of everyday purchases, while the higher average value of online transactions could reflect larger, less frequent purchases.

Customer Demographics

Gender Distribution and Banking Opportunities

Aurora Bank enjoys a near-even gender distribution, with 1,016 female customers (50.8%) and 984 male customers (49.2%). This balanced customer base offers a unique opportunity to explore gender-specific financial behaviours and tailor banking strategies accordingly.

Total Spending: Women Lead the Way

Our analysis reveals that female customers outspend their male counterparts, with a total spending of \$3,644,268.17 compared to \$3,230,215.32 for men – a 13% difference. I

Gender Distribution and Banking Opportunities: A Data-Driven Narrative

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Implications

This significant revenue disparity suggests that women are a key driver of Aurora Bank's financial performance. Understanding the reasons behind this, such as higher transaction frequency or differing spending habits, is crucial for optimizing revenue strategies.

Recommendations:

- ☐ Develop marketing campaigns that resonate with female customers, focusing on their preferred spending categories.
- ☐ Explore and enhance product offerings that cater to women's dominant spending areas, such as household essentials and personal care.

Transaction Volume: Frequency Over Value

Women not only spend more but also engage in a higher volume of transactions, recording 83,029 transactions compared to men's 74,195 – an additional 8,834 transactions. Clearly, women are engaging in more financial transactions than men.

Implications:

The trend suggests that women's higher total spending is driven by transaction frequency rather than larger individual purchases. Understanding the nature of these frequent transactions is essential for tailoring product and service offerings.

Recommendations:

- ☐ Conduct further analysis to determine if these frequent transactions represent smaller, everyday purchases or a mix of spending habits.
- ☐ Develop loyalty programs or promotional strategies that incentivize frequent transactions.

Spending Per Transaction: Minimal Difference

The average transaction value is nearly identical for both genders, with women spending \$43.89 and men \$43.54.

Implications:

Both genders tend to purchase items of similar value, with the primary difference being transaction frequency.

Recommendations:

- ☐ Focus on strategies that encourage increased transaction frequency for both genders, such as targeted promotions or rewards programs.

Spending Trends Across Categories: Navigating Gender-Specific Financial Landscapes

While women contribute a larger overall spending volume to Aurora Bank, a deeper dive into specific categories reveals distinct financial landscapes shaped by gender. Men, it appears, are drawn to the realm of high-stakes financial transactions and riskier endeavours. Notably, their spending in money transfers outstrips women's by a significant \$87,000 (\$341,960 vs \$254,600), suggesting a greater involvement in business ventures, investment opportunities, or perhaps the provision of substantial financial support. This inclination towards high-value transactions is further underscored by their \$13,000 lead in gambling and betting activities (\$38,317 vs \$25,317), indicating a higher propensity for risk-taking in pursuit of entertainment or financial gain.

Conversely, women demonstrate a clear preference for essential household and personal care purchases, as well as the management of recurring expenses. They consistently outspend men in categories like groceries (\$271,391 vs \$230,589), drug stores and pharmacies (\$237,844 vs \$192,641), and utilities (\$197,147 vs \$160,651), painting a picture of financial responsibility and a focus on maintaining household stability. This trend extends to retail and personal care, where women's spending in department stores (\$183,260 vs \$153,943) and beauty and barbershops (\$33,566 vs \$29,438) further reinforces their role in managing family-related expenditures and prioritizing self-care.

Even in dining, women's spending slightly exceeds men's, with fast food (\$84,743 vs \$74,246) and restaurant visits (\$167,730 vs \$148,305) showing their increased spending. A notable divergence, however, emerges in insurance spending, where women allocate \$28,280 more than men (\$78,532 vs \$50,252), suggesting a heightened emphasis on financial security and perhaps the purchase of more comprehensive coverage.

Implications:

These spending patterns paint a vivid picture of distinct gender-based financial priorities. Men's spending reflects a tendency towards high-value transactions and risk-taking, while women prioritize essential household needs, personal care, and financial security. This divergence presents both challenges and opportunities for Aurora Bank.

Recommendations:

- ❑ **Tailored Product Development:** Aurora should develop financial products and services that align with these gender-specific spending patterns. For men, this might include investment platforms, business loan options, and specialized financial advisory services. For women, this could involve enhanced savings accounts, family-focused financial planning, and robust insurance offerings.
- ❑ **Targeted Marketing Strategies:** Marketing campaigns should be crafted to resonate with the values and preferences of each gender, emphasizing investment opportunities and high-return financial products for men, and highlighting the security and stability of security plans and insurance products for women.

- Financial Literacy and Education: Aurora Bank should implement financial literacy programs that address the specific needs and priorities of both male and female customers. This could include workshops on investments for men and budgeting and savings tips for women.
- Risk Management and Customer
 - customers. This could include workshops on investment strategies for men and budgeting and savings tips for women.
 - **Risk Management and Customer Service:** Risk management strategies should account for the differing risk appetites of male and female customers. Customer service should be tailored to address the specific concerns and questions of each gender, ensuring a personalized and supportive banking experience.

Transaction Volume: Frequency Over Value

Women not only spend more but also engage in a higher volume of transactions, recording 83,029 transactions compared to men's 74,195—an additional 8,834 transactions.

So what? This demonstrates that women are more actively engaging with the bank's services. **The "so what" is that focusing on increasing transaction frequency among both genders could be a key strategy for growth.**

Recommendation:

- Conduct further analysis to determine the nature and frequency of female transactions (e.g., smaller, frequent purchases vs. varied spending), and tailor product offerings accordingly.

Spending Per Transaction: Minimal Difference

The average transaction value is nearly identical for both genders, with women spending \$43.89 and men \$43.54.

So what? This indicates that the difference in total spend is driven by transaction frequency, not individual transaction value. **The "so what" is that the bank should focus on strategies that encourage frequent, rather than large, transactions.**

Recommendation:

- Focus on strategies that encourage both genders to increase transaction volume, such as loyalty programs or targeted promotions.

Spending Trends Across Categories: Gender-Specific Preferences

Men dominate in money transfers and gambling-related activities, while women lead in everyday essentials, household purchases, and personal care.

So what? These distinct preferences highlight the need for tailored product offerings. **The "so what" is that the bank can significantly enhance customer satisfaction and loyalty by addressing these specific needs.**

Recommendations:

- Develop financial products and services that cater to men's interest in large financial transactions and investments.
- Focus marketing efforts on essential goods, personal care, and household-related products to target female customers.

Debt Distribution: A Slight Female Lead

Women hold a slightly higher total debt burden than men, with \$64,331,433 compared to \$63,087,955.

So what? Understanding the reasons behind this disparity is crucial for developing tailored financial products. **The "so what" is that the bank has the opportunity to provide targeted financial solutions that address the specific debt burdens of each gender.**

Recommendations:

- Tailor refinancing options and lower-interest loan products to ease the debt burden on female customers, considering their higher spending on household necessities.
- Develop targeted advisory services and investment strategies to help male customers manage their liabilities, given their tendency for high-value transactions.

Conclusion: Tailoring Strategies for Gender-Specific Success

These gender-based patterns provide valuable guidance for Aurora Bank. By tailoring strategies, the bank can enhance customer engagement, drive revenue, and foster long-term customer loyalty.

Age Distribution Analysis

Analyzing customer birth years provides invaluable insights into Aurora Bank's demographic composition, enabling a targeted approach to product development and customer engagement. The analysis identifies key age groups within the customer base, underrepresented segments (categories we think Aurora should listen more to because they are in the minority), and 'underscores the varying banking preferences across generations

The most represented birth years are 1970, 1998, 1972, and 2002. This suggests strong customer engagement from individuals in their early 20s, late 20s, late 40s, and early 50s. Notably, the presence of younger adults born in the late 1990s and early 2000s indicates a growing interest in digital banking services, a trend that Aurora Bank should pay more attention to.

A large portion of customers were born between 1965 and 1985, representing individuals likely in their prime earning years, managing mortgages, credit lines, and retirement planning. There is also a notable presence of customers born between 1920 and 1960, highlighting the need for specialized services for seniors. Finally, strong representation from customers born in 1998, 2002, 2001, and 1994 indicates active banking among those aged 22 to 26, a segment likely digitally inclined.

Implications:

- ☐ While the analysis identifies key age groups, it does not delve into the potential financial needs and challenges specific to each group. For instance, younger adults may have student debt, while pre-retirees may be concerned about healthcare costs.
- ☐ While digital inclination is noted for younger adults, the analysis does not address the digital literacy or accessibility needs of older customers.
- ☐ The analysis does not show the profitability of each group.

Recommendations:

- ☐ Conduct detailed market research to understand the specific financial needs and challenges of each age group, including debt management, investment goals, and healthcare planning.
- ☐ Develop educational resources and training programs to enhance digital literacy among older customers and ensure the accessibility of digital banking services.
- ☐ Analyze the account balances, and profitability of each age group, to better understand the financial value of each demographic.
- ☐ Create specialized financial products and services tailored to the life stage and financial priorities of each age group, such as student loan refinancing, retirement planning tools, and estate management services.
- ☐ Implement targeted marketing campaigns that resonate with the specific values and communication preferences of each age group.
- ☐ Ensure that both digital and traditional branch services are accessible and tailored to the needs of all age groups.

Income Segmentation

Income segmentation provides valuable insights into customer financial behaviour and potential banking opportunities at Aurora Bank. Majority of the customers fall within the middle-income bracket; 1,339 individuals (66.95%) earning an average annual income of \$36,537. A significant portion, 543 customers (27.15%), are classified as upper-middle-income, earning an average of \$63,820, representing a key segment for premium services. While smaller, the high-income group, totalling 52 individuals (2.6%), possesses considerable financial influence, with an average yearly income of \$135,742. At the lower end, 66 customers (3.3%) are in the low-income category, earning an average of \$12,053 annually.

Implications

This income segmentation reveals critical opportunities and potential challenges for Aurora Bank:

- ☐ The large middle-income segment represents a significant base for core banking products and services.
- ☐ The upper-middle-income segment offers potential for premium services and targeted financial products.
- ☐ Although small, the high-income segment is highly influential in the bank's overall financial health and requires specialized attention.
- ☐ The low-income segment may require tailored financial literacy programs and accessible banking solutions to foster financial inclusion.

Recommendations:

- ☐ Tailored Product Development:
 - o Develop targeted financial products and services for each income segment to maximize customer engagement and profitability.
 - o Create premium services and investment options for the upper-middle and high-income segments.
 - o Design accessible and affordable banking solutions for the low-income segment.
- ☐ Relationship Management:
 - o Provide personalized relationship management for high-income clients to maximize their value.
 - o Develop strategies to foster long-term relationships with the middle-income segment.
- ☐ Risk Assessment:
 - o Incorporate income segmentation into risk assessment models to better understand and manage credit risk
- ☐ Targeted Marketing:
 - o Implement market strategies that resonate with each income segment's specific financial needs and aspirations.
 - o Promote premium services to the upper-middle and high-income segments.

- o Focus on financial literacy and basic banking services for the low-income segment.

Segmentation by Age and Income

Analysis of spending patterns by age reveals that middle-aged customers (35-63) contribute the largest portion of total spending, amounting to \$4,618,019. This significantly surpasses the spending of both senior citizens (\$1,449,626) and youth (\$806,838).

The substantial difference in spending suggests that middle-aged individuals, typically in their peak earning years, exhibit the highest level of financial activity. Senior citizens, despite often being past their prime working years, still demonstrate considerable spending, nearly double that of youth. This could be attributed to financial stability derived from pensions, savings, or investments. Conversely, youth customers show the lowest spending, likely reflecting lower income levels or financial constraints associated with the early stages of their careers.

Implications:

- Concentrated Revenue: Middle-aged customers represent the bank's most significant revenue source, demanding tailored high-value services.
- Untapped Senior Potential: Senior citizens' substantial spending indicates a need for specialized products and services catering to their unique financial needs.
- Long-Term Youth Engagement: The lower spending of youth necessitates strategies to build long-term relationships and increase their financial engagement.
- Risk of Neglecting Segments: Ignoring the specific needs of senior and youth customers could lead to lost revenue and decreased customer loyalty.

Recommendations:

- Optimize Middle-Aged Services:
 - o Develop premium financial products and wealth management services specifically for middle-aged customers.
 - o Offer personalized financial planning and investment advice.
- Capitalize on Senior Spending:
 - o Introduce pension-friendly accounts, low-risk investment options, and healthcare-related financial

Average Spending Across Age Groups

Analysis of average transaction amounts across age groups reveals an intriguing trend. Contrary to initial expectations, the youth segment (under 35) exhibits the highest average transaction value, spending \$48.53 per transaction, followed by senior citizens at \$43.88, and middle-aged individuals at \$42.93.

So why would the data suggest that younger customers spend more per transaction compared to both middle-aged and senior customers, even as their total spend is the lowest? A potential explanation could be that youths have fewer financial obligations, allowing them to spend more on lifestyle or premium purchases.

Yet, the higher average spending of senior citizens compared to middle-aged individuals is also interesting. One might say that retirees allocate funds towards healthcare, leisure activities, and other planned expenses, while middle-aged individuals pay more attention to managing significant financial commitments such as mortgages, education costs, and family-related expenses. But these are things that could be subject to more research.

Identifying High-Risk Loan Defaulters

For banks and financial institutions to assess the financial health of a person and their tendency to default on their loans, banks use a concept called debt-to-income (DTI) ratios. Analysis carried out on the customer base identified the top ten customers facing the highest levels of debt. A higher DTI ratio suggests that a substantial portion of an individual's income is already given to debt payments, making them more prone to financial instability, and even bankruptcy.

The highest DTI ratio observed is 4.98, meaning that this individual's total debt is almost five times their annual income. This is an extreme financial burden that makes this person a potential high-risk loan defaulter.

The top three customers exhibit DTI ratios exceeding 3.9. That means that they need to earn almost 4 times their current salary at once to pay their debts. That is a severe level of financial leverage.

Even the lowest DTI ratio within this top ten list is 3.6, indicating a significantly high level of financial obligation.

Majority of these high-risk customers have incomes below \$50,000, suggesting a correlation between lower income and a greater likelihood of carrying unsustainable debt. While higher income does not guarantee immunity from debt risk, individuals with lower earnings tend to have higher DTI ratios, indicating that debt servicing poses a greater burden for them.

High-Risk Borrowers: Low Credit Scores and High Debt

After identifying the customers with the highest DTI ratios, we decided to note the top 10 customers with credit scores below 600 who also exhibit the highest debt-to-income (DTI) ratios. These individuals represent a particularly high-risk segment due to the combination of poor creditworthiness and substantial debt burdens.

The highest DTI ratio within this group is 3.9, indicating that this individual needs the equivalent of four times their annual income to pay their total debt. Even the lowest DTI ratio in the top 10 is 2.8, still an elevated level.

Credit scores within this segment range from 510 to 589, all falling well below the 600 threshold that typically separates subprime from fair credit. A majority of these customers have annual

incomes below \$50,000, compounding the challenge of debt repayment. Interestingly, customer 995, despite a higher income of \$101,679, maintains a high DTI ratio of 3.03, indicating that higher earnings do not determine whether a customer will be debt-free; poor financial management or excessive borrowing can lead to significant debt burden. Several customers (e.g., ID 1355 and ID 1017) have very low incomes (\$23,863 and \$28,703, respectively) coupled with substantial debt, placing them in a very precarious financial position.

Customers with both low credit scores and high DTI ratios represent the most vulnerable borrowers for loan default. Their limited access to traditional credit may force them to rely on high-interest or predatory loans, further exacerbating their financial challenges.

Card Usage Pattern Among Customers

Debit cards are the most prevalent among Aurora Bank's customers, with 3,511 individuals utilizing them for transactions. This suggests a preference for spending within available funds, potentially indicating a conservative approach to financial management and a possible aversion to debt. Credit card usage is notably lower, with 2,057 customers. This could be due to stricter credit approval purchases, a cultural or behavioural preference for avoiding debt, or a perception that current credit card offerings (such as rewards, cashback, or interest rates) do not adequately meet customer needs. Prepaid debit cards are the least utilized, with only 578 users. This may indicate their primary use for budgeting or controlled spending, potentially catering to lower-income customers or those lacking access to traditional banking services.

On average, each customer holds three credit cards, indicating a tendency to maintain multiple credit lines. This could reflect a desire for diversified credit options, a strategy to distribute spending across different cards, or an effort to maximize benefits from rewards and cashback programs. However, this also raises important considerations regarding potential over-extension of credit and debt management, particularly if customers are utilizing multiple cards as a means of coping with financial strain rather than for strategic financial planning.

Average Credit Card Holdings by Age Group

Senior citizens hold the highest average number of credit cards at four per customer, followed by middle-aged individuals with three, while younger customers hold the least, averaging two. This suggests that older customers have likely accumulated more credit lines over time, potentially due to stronger credit histories, greater financial needs, or a preference for diversifying their spending. Middle-aged individuals, often in their peak earning years, maintain a moderate number of cards, balancing financial flexibility with responsible credit management. Younger customers, who are typically early in their financial journey, hold fewer cards, likely due to stricter credit approval processes or a preference for simpler financial management.

Number of Cards Held by Customers

The majority of customers (3,114) hold only one card, suggesting a preference for single-card usage or a lack of perceived need for multiple cards. A substantial segment (2,972) holds two cards, indicating a significant portion of customers prefer to diversify their payment options, potentially balancing their use of credit and debit cards.

Only 60 customers hold three cards, representing a small minority. This could suggest that holding multiple cards is not a common practice among the customer base, potentially due to financial management strategies, credit limits, or spending habits.

Card Type Distribution Among Customers

An overview of card type distribution among our 2,000 customers reveals that 1,313 have credit cards, 1,702 have debit cards, and 509 have prepaid debit cards. Notably, the number of customers holding both a credit card and at least one other card type is also 1,313. This indicates that all credit card holders also possess either a debit or prepaid card. The data suggests that debit cards are the most common, followed by credit cards, with prepaid cards showing the lowest adoption rate.

Distribution of Card Brands Among Customers

Mastercard is the most prevalent card brand among Aurora Bank's customers, with 3,209 cards issued, surpassing Visa, which has 2,326 cards in circulation. This suggests that Mastercard may have more favourable partnerships, incentives, or acceptance rates among customers.

American Express (Amex) and Discover exhibit significantly lower adoption, with 402 and 209 cards, respectively. This could indicate that these brands are less preferred due to their fees, acceptance limitations, or specific customer demographics that favour other options.

The dominance of Mastercard and Visa aligns with their global reputation for accessibility and widespread merchant acceptance, reinforcing their status as the primary choices for most customers.

Card Type Usage Across Age Segments

Analysis reveals that middle-aged customers constitute the largest segment across all card types, with 1,847 using debit cards and 1,102 using credit cards. Senior citizens and youth exhibit a smaller distribution in debit card usage, with 840 and 824 users, respectively. However, credit card adoption is lower among both groups, with youth at 492 and senior citizens at 463. Prepaid debit cards show the lowest usage across all age segments, with middle-aged customers leading at 294, followed by youth at 150 and senior citizens at 134. This distribution suggests that debit cards are the most widely used across all age segments, while credit card adoption is notably higher among middle-aged individuals.

Card Expirations in 2024

Card expirations in 2024 totalled 962, signalling a potential disruption in customer transactions if replacements or renewals have not been secured. This volume of expirations introduces multiple implications, including temporary lapses in spending, shifts to newly issued cards, and possible changes in consumer behaviour. Businesses reliant on recurring payments may face an uptick in declined transactions, necessitating proactive customer outreach to mitigate revenue impact. Additionally, spending patterns may temporarily fluctuate as customers transition to updated

payment methods, potentially influencing short-term revenue trends across various merchant categories.

Prevalence of Chip-Enabled Cards

A significant majority of cards, totalling 5,500, are equipped with chips, while a smaller portion, 646, are not. This indicates that most customers are utilizing more secure, chip-enabled cards, which offer enhanced protection against fraud compared to traditional magnetic stripe cards. The presence of non-chip cards suggests that some customers may still be using older card technology, potentially exposing them to higher security risks and limited acceptance in certain transactions. Monitoring the phasing out of these non-chip cards over time could provide valuable insights into the progress of security upgrades and the rate of customer adoption of newer payment technologies, a work beyond the scope of this project.

Customers with Stagnant PINs

Our findings revealed that 4,271 customers were yet to change their PINs in 10 years. This poses a noteworthy security concern. Older PINs are inherently more susceptible to theft or unauthorized access, thereby elevating the risk of fraudulent activity. This situation could stem from customer inactivity, a lack of awareness regarding the risks, or insufficient enforcement of PIN change policies by the bank.

It might be required to conduct further analysis on whether these customers are still active and whether old PINs correlate with security incidents, a work beyond the scope of the project.

Spending Patterns by Merchant Category

Analysis of transaction data reveals that essential goods and services constitute the majority of customer spending. Grocery stores and supermarkets lead with 18,371 transactions, closely followed by miscellaneous food stores (17,969 transactions) and service stations (17,055 transactions). This indicates that a significant portion of consumer spending is directed towards daily necessities such as food, fuel, and basic utilities.

Dining experiences also play a substantial role in spending patterns, with restaurants and eating places accounting for over 11,673 transactions. The 5,888 transactions at fast food restaurants further emphasize the frequency of food-related expenditures.

Beyond essentials, financial transactions like money transfers (7,257 transactions) and retail shopping at department stores (5,938 transactions) highlight diverse spending behaviour. However, discretionary spending categories such as theatres (1,603 transactions), amusement parks (140 transactions), and gift shops (31 transactions) show significantly lower transaction volumes. This suggests that customers allocate less spending towards entertainment and luxury purchases compared to necessities.

Expenditures on medical services, insurance, and professional services (e.g., legal, tax, and accounting services) are also relatively low, potentially indicating that these categories involve larger but less frequent payments.

Overall, the spending distribution underscores a consumer focus on day-to-day essentials and moderate discretionary spending, with food, transportation, and household needs taking precedence over luxury and leisure expenses.

Total Transaction Value by Merchant Category

Money transfers account for the highest total transaction value across all merchant categories, indicating a significant volume of fund transfers among customers. Grocery stores and supermarkets follow closely, reflecting consistent spending on daily necessities. Wholesale clubs, pharmacies, and service stations also demonstrate substantial transaction values, suggesting regular expenditures on household essentials, health needs, and fuel. Categories such as sporting goods, gift stores, and cosmetic stores record considerably lower transaction values, implying less frequent purchases or lower overall spending in these areas.

So what? This spending pattern provides valuable insights into customer priorities and needs. The high volume of money transfers suggests a potential demand for enhanced money transfer services or products. The consistent spending on essentials highlights the importance of partnerships or offerings related to these categories. The lower spending in discretionary areas can inform marketing strategies and product development, potentially leading to targeted promotions or loyalty programs to encourage spending in these areas. Understanding these spending patterns allows Aurora Bank to tailor its services and offerings to better meet the everyday needs and financial behaviours of its customer base.

Contrasting Transaction Values Across Industries

The data reveals a striking contrast between industries characterized by frequent, low-value transactions and those defined by large, less frequent purchases. Money transfers, while leading in total transaction value, exhibit an average transaction amount of \$82, reinforcing that their dominance stems from volume rather than individual spending. Similarly, grocery stores and supermarkets, with an average transaction amount of just \$27, confirm their role as high-frequency, essential spending categories.

Conversely, cruise lines command the highest average transaction at \$1,188, followed by manufacturing-related industries like tools, parts, and steelworks, where transactions often exceed \$700. This indicates that while these industries process fewer transactions overall, each purchase carries significant financial weight. Airlines also rank high in average transaction value at \$746, highlighting the costly but less frequent nature of travel-related expenses.

Wholesale clubs and pharmacies, despite their high total transaction values, show moderate average transaction amounts at \$62 and \$48, respectively. This suggests a pattern of bulk purchasing, where customers spend more per transaction but shop less frequently than at grocery stores. Service stations, representing routine spending, register an average transaction of just \$22, reinforcing the frequent but low-cost nature of fuel purchases.

Luxury and niche markets such as jewelry, boutique stores, and specialty retailers do not prominently feature in either total value or average transaction size, suggesting a reliance on selective, high-margin sales rather than widespread consumer spending. Entertainment-related industries, including movie theatres and sporting goods stores, fall into a discretionary spending category, characterized by transactions that are lower in both frequency and value compared to everyday necessities.

This analysis underscores distinct consumer spending patterns: necessities like groceries, fuel, and healthcare services dominate in frequency, while travel, manufacturing, and luxury purchases emerge as high-value but less common transactions.

So what? This information is valuable for Aurora Bank in several ways. Understanding these spending patterns across different industries can inform:

- ❑ Partnership Opportunities: Identifying industries with high transaction volume (e.g., grocery stores, service stations) can lead to potential partnership opportunities for rewards programs or co-branded cards.
- ❑ Credit Card Rewards Design: Recognizing industries with high average transaction values (e.g., travel) allows for the design of targeted rewards program that incentivize spending in these areas.
- ❑ Risk Assessment: Understanding spending patterns in discretionary categories can help in assessing credit risk and potential for default.
- ❑ Customer Segmentation: These insights can contribute to more refined customer segmentation based on spending habits and preferences, enabling more targeted marketing and product offerings.

Transaction Value by Location

The transaction data reveals that California leads all states in total transaction value, reaching approximately \$778,631.78. Texas follows with \$666,418.28, and New York ranks third at \$459,882.16. These figures suggest that these states experience the highest consumer spending, likely due to their large populations and economic activity. Florida and North Carolina also show significant transaction volumes, reinforcing the trend that major economic hubs and densely populated states drive higher financial activity.

Interestingly, a handful of international transactions appear in the dataset. Italy, Mexico, and Canada record notable amounts, suggesting cross-border payments or international commerce. However, some countries, such as Norway and Spain, report negative transaction values, potentially indicating chargebacks, currency adjustments, or fraud-related reversals.

Additionally, Wyoming, at \$117.91, records the lowest domestic total, reflecting its small population and lower transaction volume. The presence of negative values for certain locations requires further investigation to determine whether they result from fraud, reporting errors, or business-specific accounting adjustments.

So what? This geographic distribution of transaction value provides Aurora Bank with valuable insights for strategic planning:

- ❑ **Market Focus:** The concentration of high transaction value in states like California, Texas, and New York can inform decisions about resource allocation, marketing efforts, and potential branch locations.
- ❑ **International Opportunities:** The presence of international transactions, particularly in Italy, Mexico, and Canada, suggests potential opportunities for expanding services related to cross-border payments and international commerce.
- ❑ **Risk Management:** The negative transaction values in certain countries necessitate a deeper investigation to understand the underlying causes and potentially implement measures to mitigate fraud or reporting errors.
- ❑ **Targeted Marketing:** Understanding regional spending patterns can enable more targeted marketing campaigns and product offerings tailored to the specific needs and behaviours of customers in different geographic areas.

Average Transaction Value by Location

The average transaction amount by merchant location presents an interesting contrast to the total transaction values. Notably, Denmark (\$166.83) and Russia (\$161.79) lead the list, indicating that while the total number of transactions in these locations might be lower, individual transactions tend to be of higher value. This pattern is consistent with luxury purchases, specialized services, or high-value transactions prevalent in these countries.

Several other international locations, including Hong Kong (\$133.29), Finland (\$105.01), and Singapore (\$94.82), also rank highly, suggesting that these markets see fewer but more substantial transactions, possibly driven by business expenditures or high-cost consumer goods.

Among U.S states, Rhode Island (\$84.74) and Alaska (\$68.25) have the highest average transaction values, which could be attributed to the higher cost of goods and services in these regions. Conversely, major economic states like California (\$40.47) and Texas (\$44.46) fall closer to the middle, reflecting a high volume of smaller transactions.

At the lower end, Wyoming registers an average transaction value of just \$4.07, indicating a predominance of small-ticket purchases. The presence of negative values for some countries, such as Switzerland (-\$190.54) and Portugal (-\$222.51), suggests potential chargebacks, fraud adjustments, or currency conversion discrepancies. These causes warrant further investigation to determine their underlying causes and whether they reflect systemic issues or isolated incidents.

So what? Analysing the average transaction value by location provides Aurora Bank with insights that complement the total transaction value analysis and can inform several strategic decisions:

- ❑ **Understanding Market Dynamics:** The high average transaction values in countries like Denmark and Russia suggest that while these markets may have lower transaction volumes, they represent opportunities for targeting high-value customers or specialized financial services.

- ❑ Geographic Pricing Strategies: The higher average transaction values in states like Rhode Island and Alaska could indicate a need to consider regional pricing difference or the promotion of products and services aligned with higher costs of living.
- ❑ Risk Management and Fraud Detection: The negative values in certain countries necessitate a focused investigation into potential fraud or chargeback patterns, allowing for the development of more effective fraud prevention strategies.
- ❑ Cross-Border Payment Solutions: The data highlights the importance of efficient and cost-effective cross-border payment solutions, particularly in regions with high average transaction values or significant total transaction value.

Failed Transactions

Our findings reveal a total of 3,737 failed transactions attributed to various errors. Understanding the causes of these failures is crucial for enhancing transaction success rates, minimizing customer dissatisfaction, and addressing potential security vulnerabilities.

Dominant Causes of Failure:

The most prevalent reason for failure is “Insufficient Balance” (1,760 cases), accounting for nearly half of all failed transactions. This suggests that a significant portion of customers attempt transactions without enough money, potentially due to a lack of real-time balance awareness or delays in updating available funds.

The second most frequent reason is “Bad PIN” (388 cases), indicating that customers may be entering incorrect PINs or that there could be issues with card authentication processes. Repeated failed attempts concentrated around specific users may signal potential fraud attempts or forgotten credentials.

“Technical Glitch” (324 cases) ranks third among failure reasons. This could be due to network outages, banking system downtime, or integration issues between transaction platforms and payment processors.

Card Authentication Errors:

Failures related to incorrect card details, including “Bad Card Number” (93 cases), “Bad Expiration” (73 cases), “Bad CVV” (65 cases), and “Bad Zipcode” (17 cases), collectively represent 6.5% of all failed transactions. These errors may indicate manual entry mistakes by customers, expired or outdated card information, or potential fraudulent attempts involving repeated testing of incorrect card details.

Multiple Error Cases:

A small number of transactions failed due to multiple simultaneous errors (e.g., “Bad PIN & Insufficient Balance” or “Bad Card Number & CVV”). These cases highlight complex transaction failures where more than one issue contributes to the rejection.

So what? Analyzing these failed transactions provides Aurora Bank with critical insights for improving operational efficiency, enhancing customer experience, and strengthening security protocols:

- ❑ Customer Education: The high number of “Insufficient Balance” failures suggest a need for improved customer education regarding balance awareness and real-time updates.
- ❑ Security Enhancement: The “Bad PIN” failures, especially repeated attempts, necessitate a review of card authentication processes and the potential implementation of measures to detect and prevent fraud.
- ❑ System Reliability: The “Technical Glitch” failures highlight the importance of maintaining robust and reliable banking systems and network infrastructure.
- ❑ User Experience Improvement: Addressing card authentication errors can improve the overall user experience by reducing friction during transactions.
- ❑ Fraud Prevention: Investigating the “Multiple Error Cases” and the card authentication failures can provide valuable insights into potential fraud patterns and inform the development of more effective fraud prevention strategies.

Financial Impact of Failed Transactions

The primary financial loss from failed transactions stems from insufficient balance, totalling \$123,259.26 This indicates a significant issue with customers attempting transactions with inadequate funds. Technical glitches are the next major contributor, resulting in a \$15,212.79 loss, suggesting potential system-related failures impacting both user experience and merchant transactions. Bad PIN errors follow, with a \$14,874.17 loss, highlighting potential user errors or fraudulent attempts.

Bad card number and bad CVV errors contributed \$5,826.96 and \$3,599.16 in losses, respectively, suggesting a mix of user input mistakes and potential fraud. Bad expiration and bad zipcode errors contributed \$3,485.01 and \$527.95, indicating issues with outdated card information or incorrect address details.

Combined errors, such as bad PIN and insufficient balance (\$384.34) or insufficient balance and technical glitches (\$234,39), also contributed to losses, highlighting recurring issues.

The prevalence of failures due to insufficient funds suggests a need for improved balance awareness tools. Technical glitches require investigation to identify patterns affecting specific merchants or transaction types. High-value failures due to authentication errors (PIN, CVV, and number) suggest potential fraud. Further analysis of these patterns can identify accounts or merchants with repeated failures. Addressing validation measures and system reliability are crucial to reducing the total amount lost due to failed transactions.

So what? The financial impact of these failed transactions directly affects Aurora Bank’s profitability and operational efficiency. The significant losses due to insufficient funds highlight an opportunity to improve customer communication and potentially offer tools for better balance management. The losses from technical glitches point to a need for investment in system upgrades and maintenance to ensure reliability and minimize disruptions. Investigating and

addressing the authentication errors can lead to enhanced fraud prevention measures, protecting both the bank and its customers from financial losses. Ultimately, reducing failed transactions will improve customer satisfaction, decrease operational costs associated with handling errors, and strengthen the bank's financial performance.

Failed Transactions by Location

California exhibits the highest amount due to insufficient balance (\$14,579.36), followed by Texas (\$10,719.70) and Florida (\$9,742.39). New York and North Carolina also show significant losses from this cause. Notably, "NULL" entries in the dataset, amounting to \$10,387.81, indicate failed transactions not associated with a specific state, potentially representing international or unclassified activity.

Technical glitches also contribute to failures, with California experiencing the highest losses (\$2,355.70), followed by Texas and Georgia. These failures suggest potential processing errors, system downtime, or issues with payment gateways.

Bad PIN errors

Bad PIN errors are prominent in Texas, Italy, and California, suggesting frequent user mistakes or potential fraud attempts. Bad card number and bad CVV failures also occur frequently, with unclassified locations accounting for the largest losses.

The data indicates that insufficient balance is the primary reason for failed transactions across multiple states. However, technical glitches and user input errors are also significant contributors. The presence of international locations suggests that transaction failures are not limited to the U.S., indicating the need for improved global payment verification.

Implications

- ❑ **Regional Disparities in Financial Awareness:** The concentration of insufficient balance failures in specific states like California, Texas, and Florida suggests varying levels of financial awareness or account management practices among customers in different regions.
- ❑ **Technical Infrastructure Vulnerabilities:** The significant losses from technical glitches, particularly in California, Texas, and Georgia, point to potential weaknesses in the bank's technical infrastructure or payment processing systems.
- ❑ **Geographic Variations in Fraud Risk:** The prominence of bad PIN errors in certain locations, including international ones, suggests that fraud risk may vary geographically, necessitating tailored security measures.
- ❑ **Global Payment Verification Challenge:** The presence of international transactions and failures underscores the complexity of global payment verification and the potential for cross-border transaction issues.
- ❑ **Operational Inefficiencies:** The patterns of recurring failures, particularly technical issues and high-risk areas, indicate potential operational inefficiencies that may be impacting transaction success rates.

Leveraging these insights will empower Aurora Bank to make informed decisions to improve its services, enhance security, and optimize its operations, ultimately leading to a better customer experience and stronger financial performance.

Failed Online Transactions: Patterns and Implications

Our findings revealed that failed online transactions involving a merchant state called “NULL” exceeded \$26,000 in losses. That caught our attention, because such transactions could be put as international transactions or as unclassified activity. Further investigations revealed that these transactions were online transactions because they had the merchant city to be “ONLINE.” After seeing that, the data became clearer.

The data reveals \$26,353.81 in failed online transactions, highlighting unique challenges in digital payments. Insufficient balance is the leading cause, exceeding \$10,000 in losses, indicating customers attempting purchases beyond their means. Bad card number errors follow, suggesting user mistakes or potential fraud. Technical glitches, accounting for over \$2,600 in failures, point to possible system inefficiencies.

Unlike in-person transactions, online payments face barriers related to authentication, trust, and system stability. Technical glitches suggest failures stem from both user behaviour and digital payment infrastructure. The question arises whether online transaction errors are disproportionately higher than in-person failures, indicating a potential gap in handling digital versus physical payments. It also raises concerns about demographic differences in online payment struggles and repeat failure patterns.

These findings suggest failed online transactions follow distinct patterns, offering opportunities to refine the online payment experience. Addressing financial constraints, improving input accuracy, and optimizing systems can mitigate failures.

Implications

The high incidence of insufficient balance failures indicates a need to address customer financial awareness in the online context. Bad card number errors suggest potential vulnerabilities in online security or user input processes. Technical glitches point to potential system reliability issues specific to online transactions. The disparity between online and in-person failures highlights a potential gap in service delivery. Further investigation into demographic and repeat failure patterns can provide insights into specific customer needs and behaviours in the digital space, a project beyond the scope of this project.

Failed Online Transaction Volume

In two years, a total of 445 online transactions failed due to errors. Numbers like this suggest that these failures are not isolated incidents but a recurring issue impacting a significant number of transactions. With a total failed transaction amount of \$26,353.81, the average failed transaction represents approximately \$59.

This pattern indicates a substantial portion of customers experience difficulties completing online payments. The association of these failures with identifiable errors, such as insufficient funds and technical glitches, suggests potential systemic challenges in digital payment processes rather than solely individual user errors.

The occurrence of 445 failed transactions also raises concerns about potential revenue loss and customer frustration. Repeated failures could impact user trust in online payments, potentially leading to cart abandonment or a shift to alternative payment methods.

Implications

- ❑ **Customer Expectations:** In today's digital age, customers expect seamless and reliable online transactions. Even a small number of failures can erode trust and negatively impact the bank's reputation.
- ❑ **Lost Revenue:** Each failed transaction represents potential lost revenue for merchants and, indirectly, for the bank through lost transaction fees.
- ❑ **Severe Negative Customer Experience:** The high volume of failures suggests a significantly negative customer experience, leading to frustration, annoyance, and potential loss of customer loyalty.
- ❑ **Potential Customer Trust Erosion:** The frequency of failed transactions raises concerns about a potential erosion of customer trust in the reliability of online payment systems.
- ❑ **Potential for Growth:** Online transactions are a growing segment of the market. Any significant number of failures can hinder the bank's ability to capitalize on this growth.
- ❑ **Security Concerns:** Failed transactions can sometimes indicate underlying security vulnerabilities or fraudulent activity, which need to be addressed promptly.
- ❑ **Systemic Issues:** The sheer number of failed online transactions points to potential systemic challenges within the digital payment infrastructure, impacting a broad customer base.
- ❑ **Competitive Disadvantage:** If other banks offer a more reliable online transaction experience, customers may switch to their competitors.
- ❑ **Need for Targeted User Analysis:** The necessity to analyze if these failed transactions are concentrated among a specific group of users or spread evenly across different segments. Identifying patterns in transaction failure rates could provide deeper insights into who is most affected and why.
- ❑ **Potential for High Cart Abandonment:** The data suggest a potential for high cart abandonment due to payment error.

Overall Transaction Failure Rate Analysis

Of 157,224 total transactions, 154,486 were successful, while 2,738 encountered errors, resulting in a failure rate of approximately 1.74%. While seemingly low, this percentage represents a significant volume of failed payments due to the overall transaction scale.

Implications

These failures have implications beyond the numbers. Each unsuccessful transaction does the following:

- ❑ Significant Transaction Disruption: Even a seemingly low failure rate of 1.74% represents a substantial disruption given the large volume of transactions.
- ❑ Customer Experience Degradation: Failed transactions lead to customer frustration and potential loss of loyalty, impacting the overall customer experience.
- ❑ Revenue Loss of Merchants: Merchants experience direct revenue loss from failed transactions, affecting their business operations.
- ❑ Increased Operational Costs: The need for customer support and refund processes associated with failed transactions increases operational costs.
- ❑ Systemic Vulnerabilities: Technical glitches suggest underlying systemic vulnerabilities in the transaction processing infrastructure.
- ❑ Customer Education Gaps: User-related errors highlight potential gaps in customer education or the need for more user-friendly payment options.
- ❑ Online Transaction: The disproportional contribution of online transactions to the failure rate indicates a need for focused attention on improving online payment processes.

Failed Transactions by Transaction Method

Of the 2,738 failed transactions, chip transactions accounted for the highest number of errors, with 1,818 failures. This significantly exceeds both swipe transactions (475) and online transactions (445).

The high failure count of chip transactions, despite their enhanced security design, raises questions about potential technical malfunctions at payment terminals, chip-reading errors, or customer-related issues like improper card insertion. The prevalence of chip transaction failures in in-person purchases suggests possible location-based or merchant-specific trends.

Swipe transactions, while fewer than chip transactions, still recorded 475 errors. Given the security vulnerabilities and the shift away from swipe-based payments, these errors warrant monitoring, particularly if certain segments or merchants rely heavily on swiping.

Online transactions recorded 445 failures, the lowest among the three categories. These errors might stem from customer input mistakes or security measures, given online transactions' susceptibility to card-related issues.

Implications:

- ❑ Chip Transaction Reliability: The disproportionately high failure rate of chip transactions indicates potential reliability issues with chip technology or its implementation in payment terminals.
- ❑ In-Person Transaction Vulnerabilities: The concentration of chip transaction failures in in-person purchases suggests potential vulnerabilities at specific merchant locations or with certain payment terminals.

- ❑ **Swipe Transaction Security Concerns:** The continued presence of swipe transaction failures, despite security concerns, indicates that certain customer segments or merchants may still be relying on this less secure method.
- ❑ **Online Transaction Authentication Challenges:** Online transaction failures suggest potential challenges related to customer authentication, card verification, or fraud detection.
- ❑ **Discrepancy Between Security and Reliability:** The fact that the most secure transaction method (chip) has the highest failure rate highlights a potential discrepancy between security and reliability, raising questions about the overall effectiveness of the current payment infrastructure.
- ❑ **Merchant Training and Terminal Maintenance:** The high chip failure rate could indicate a need for enhanced merchant training on proper chip usage and regular maintenance of payment terminals.
- ❑ **Customer Education on Card Usage.** Customer related issues with chip usage also points to a need for better education on proper card insertion.

Financial Impact of Failed Transactions by Transaction Method

Chip transactions, with the highest error count, also resulted in the greatest lost transaction value—over \$111,335. This is nearly four times the loss from swipe transactions (\$30,089) and online transactions (\$26,353).

The high monetary loss in chip transactions reinforces that failures in this category are not only frequent but involve substantial transaction values. This suggests chip transactions are predominantly used for higher-value purchases, making each failure financially significant.

Swipe transactions, despite being less secure, had a lower financial impact. However, online transactions, while having the fewest failures, still accounted for \$26,353 in lost transaction value. This aligns with findings that online transactions often fail due to card-related issues.

Implications

- ❑ **High-Value Chip Transaction Vulnerability:** The significant financial loss from chip transaction failures indicates a vulnerability in high-value-in-person purchases.
- ❑ **Disproportionate Financial Risk:** The substantial difference in loss between chip and other transaction types highlights a disproportionate financial risk associated with chip transactions.
- ❑ **Chip Transaction Usage Patterns:** The high loss suggests chip transactions are predominantly used for higher-value purchases, informing targeted interventions.
- ❑ **Swipe Transaction Value Differences:** The lower financial impact of swipe transactions suggests potential differences in usage patterns and average transaction compared to chip transactions.
- ❑ **Merchant and Issuer Vulnerabilities:** The recurring issues with chip transactions suggest potential vulnerabilities at the merchant or card issuer level.
- ❑ **Need for Targeted Investigation:** The financial impact necessitates a deeper dive into merchant-specific trends and error types to uncover actionable insights.

Failed Transactions by Merchant

We ran a query to look at which merchants were having the highest failed transactions. To do this, we set the scale for merchants having more than 100 errors.

Our findings revealed that Merchant ID 27092 stands out with the highest failure of 3.11%, indicating that out of 7,257 total transactions, 226 failed. This significant deviation from the overall failure rate suggests a potentially serious issue specific to this merchant. Following this, Merchant ID 39021 has a failure rate of 2.38%, with 139 failed transactions out of 5,828 total. Merchant ID 59935 shows a failure rate of 1.96%, with 144 failures out of 7316 total transactions. Lastly, Merchant ID 61195 has the lowest failure rate among the displayed results, at 1.56%, with 113 failures out of 7228 total transactions.

Implications

These results reveal a notable variation in failure rates among merchants, indicating that the causes of failures are likely merchant-specific. Merchant 27092 is a clear outlier, suggesting a potentially severe or unique problem at this location. The fact that all these merchants have over 100 failed transactions indicates that these are not isolated incidents but recurring problems. These failure rates can lead to lost revenue for merchants, customer frustration and dissatisfaction, increased operational costs for both merchants and the bank, and potential security risks. To gain a deeper understanding of these failures, further analysis should include examining the specific error codes associated with these failed transactions to identify the most common causes, analysing the timing of the failures to see if there are any patterns related to specific days or times, determining the types of transactions that are failing to identify potential points of failure. This detailed analysis will provide valuable insights into the root causes of these failures. However, it is not in our project scope.

Failure Rates by Merchant State

Florida (FL) exhibits the highest failure rate at 2.54% among the listed states. This indicates a potential concentration of transaction-related problems within Florida, such as heightened fraud risk or processing inefficiencies.

As we have seen, the “NULL” category, representing online transactions, displays a failure rate of 2.48% with 445 failed transactions. It is a significant concern as it reveals a substantial volume of failed transactions from merchants with missing location data. The absence of state information for these transactions warrants immediate investigation to understand the root cause.

North Carolina (NC) follows with a 2.23% failure rate, while New York (NY), Texas (TX) and California (CA) have failure rates ranging from approximately 1.5% to 1.66%.

California (CA), despite having the highest total transaction volume (19,241), demonstrates a relatively low failure rate of 1.54% compared to Florida. This suggests a more stable transaction processing environment in California.

Implications:

- ❑ Regional Transaction Inefficiencies: The high failure rate in Florida points to potential regional inefficiencies in transaction processing or elevated risk factors.
- ❑ Online Transaction Vulnerabilities: The near-highest failure rate associated with online (“NULL”) transactions highlights potential vulnerabilities in the bank’s online payment infrastructure or security protocols.
- ❑ Data Integrity Concerns: The “NULL” category also raises concerns about data integrity, as the absence of state information for a significant number of transactions indicates potential issues with data collection or processing.
- ❑ Geographic Fraud Risk Variation: The differences in failure rates across states suggest that fraud risk may vary geographically, requiring tailored fraud prevention strategies.
- ❑ Transaction Volume vs Stability: The comparison between California and Florida demonstrates that high transaction volume does not necessarily equate to high failure rates, highlighting the importance of robust transaction processing systems.
- ❑ Focus on Online Security: The high “NULL” rate, being online, shows that online security, and or customer experience, needs to be looked at.
- ❑ The need to better classify online transactions; The “NULL” state should be investigated to find out why online transactions are not being classified correctly and bringing out the merchant location.

Fraud Detection

Repeated Failed Transactions Analysis

Over the past two years, transaction data reveals a concerning pattern of repeating failed transactions, with some clients experiencing dozens of errors. Client 464 exhibits the highest failure count at 55, followed by client 373 with 42. Numerous other clients face over 30 unsuccessful attempts, indicating a widespread issue.

These repeated failures, particularly for those with 20 or more attempts, suggest persistent challenges beyond occasional inconveniences. Potential causes include payment processing failures, system errors, fraud detection blocks, or user mistakes. The impact extends to lost revenue, increased operational costs, and the risk of customer attrition.

The scale of these failures raises questions about the underlying causes. Potential factors include technical glitches, repeated user errors, or systemic issues within the payment processing system.

Implications:

- ❑ Systemic Transaction Reliability Issues: The sheer volume of repeated failures indicates potential systemic issues within the bank’s transaction processing infrastructure.

- ❑ Customer Experience Deterioration: Repeated failures lead to significant customer frustration and potential loss of trust in the bank's services.
- ❑ Revenue Leakage and Increased Operational Costs: Failed transactions result in lost revenue and increased operational costs associated with customer support and error resolution.
- ❑ Potential for Customer Attrition: The high frequency of failures increases the risk of customers abandoning transactions and switching to competitors.
- ❑ Fraud Detection Algorithm Inefficiencies: If fraud prevention measures are mistakenly blocking legitimate transactions, this points to inefficiencies in the fraud detection algorithms.
- ❑ Customer Support Gaps: The fact that customers are experiencing repeated failures without apparent resolution indicates potential gaps in customer support or proactive engagement.
- ❑ User Error Patterns: Repeated errors could point to user error patterns that need to be addressed with better customer education.
- ❑ Need for Proactive Monitoring and Intervention: The data highlights the need for proactive monitoring of transaction failures and timely intervention to address underlying issues.

Repeated Failed Transactions: Potential Fraud Indicators

While inconclusive, the frequency of repeated failed transactions raises serious concerns about potential fraudulent activity. For example, Client 464's 55 failures within two years significantly deviate from typical user behaviour. Despite these repeated failures, the client remains active, with a last transaction date of October 31, 2024, suggesting continued platform interaction. This pattern raises suspicion, as it is unlikely a legitimate user would tolerate such persistent issues.

Further analysis of last transaction dates reveals a trend: many clients with high failure rates remains active, raising the possibility of deliberate attempts to bypass security measures.

Conversely, some high-failure clients have been inactive for extended periods. Clients 456 (22 failures, last transaction June 9, 2024), 86, 291, and 442 exhibit numerous failed attempts despite prolonged inactivity. This strongly suggests potential unauthorized access or fraudulent activity.

It is possible that fraudsters are using these accounts to attempt numerous transactions to validate stolen card details, or gaining account access through leaked login details. In addition, bots might be used to rapidly test payment details and exploit system vulnerabilities. But this is inconclusive and should be looked into.

Implications

For Aurora Bank, they have to make immediate security enhancements for the following reasons:

- ❑ Increased Fraud Risk: Inactive accounts with failed attempts are prime targets for unauthorized access and fraud.

- ❑ Potential Credential Compromise: Active accounts with persistent failures suggest user confusion, card testing, or potential credential compromise, all of which require immediate action.
- ❑ Reputational Damage: Unresolved security issues can erode customer trust and damage the platform's reputation.
- ❑ Financial Losses: Fraudulent transactions and increased support costs associated with resolving these issues can lead to substantial financial losses.

Unusual Spending: Potential Fraud Indicators

To gain a better understanding of the potential fraud lurking, we looked at the average and maximum transaction amounts for each client and focused on clients whose maximum transaction exceeded five times their average transaction.

Analysis revealed the presence of clients with transactions significantly larger than their usual spending habits. This suggests a variety of potential explanations, including occasional large purchases or investments, unpredictable spending patterns, or possible errors such as accidental inputs, refunds, or bulk purchases. The data indicates a wide range of spending behaviours, from moderate to extreme outliers. For instance, Client 278 demonstrate a maximum transaction of \$1,911.54, while their average spending was only \$75.88, indicating a highly irregular transaction. In contrast, Client 349 showed a maximum transaction of \$573.74, with an average of \$19.60, still exceeding five times their usual spending.

The distribution of these outliers showed that the top 10 clients had maximum transactions exceeding \$1,500. Additionally, a significant number of clients had transactions between \$600 and \$1,200, suggesting this could be a common range for substantial purchases. Clients whose average spending was below \$50 but whose maximum spending exceeded \$1,000 exhibited the most drastic jumps in transaction size.

The implications depend on the perspective one uses. If they are instances of occasional luxury or one-time high-value purchases, then it could be an opportunity for targeted marketing. But it may also signal potential fraud or input errors, necessitating enhanced fraud detection measures.

Unusual Transaction Hours Analysis

As part of our fraud detection, we looked for clients who made transactions at unusual hours (midnight to 5 AM). Our findings reveal unusual activity patterns, with several clients conducting transactions during late-night hours (midnight to 5 AM). Client 114 exhibits the most significant activity, with 527 late-night transactions, followed by Clients 301 (263 transactions) and 121 (236 transactions), indicating habitual late-night spending.

Clients previously flagged for suspicious activity, such as 394 (179 transactions) and 464 (153 transactions), also demonstrate significant late-night transaction volumes, further raising concerns about their financial behaviours. Even Client 17, with 85 late-night transactions warrants closer scrutiny.

Implications

- Potential for Irregular Business Operations: The high volume of late-night transactions suggests potential involvement in businesses that operate outside of standard hours, which could be legitimate but also requires verification.
- Possible Money Laundering Activities: Late-night transactions could be used to obscure the movement of funds, potentially indicating money laundering activities.
- Increased Risk of Fraudulent Activity: Unusual transaction hours can be a red flag for fraudulent activity, as fraudsters often operate during periods of reduced monitoring.
- Need for Enhanced Transaction Monitoring: The identified patterns necessitate enhancing monitoring of late-night transactions to identify and mitigate potential risks.
- Potential for Concealed Activities: Late night transactions could be an attempt to conceal activities from normal banking oversight.
- Need to cross reference with other anomalous data: The late-night transaction data needs to be cross-referenced with the other data points that have shown anomalous behaviour.

Observations on Client Transactional Anomalies

During our review, several client transaction patterns have diverged from expected norms, warranting closer scrutiny. While many clients exhibit standard transactional behaviour, others demonstrate activity that raises pertinent questions.

Client 464, notably, displays significant financial activity. The transfer of \$35,080, coupled with substantial expenditures at automotive service shops summing up to nearly \$29,000, and a series of unusual refunds across diverse sectors, suggests a pattern that merits further investigation. The possibility of money cycling or rapid purchase reversals is a point of interest.

Client 96 also presents anomalies. While their money transfers are less substantial at \$4,500, the significant spending at service stations, amounting to \$23,604, and the presence of scattered, unusual refunds in specialized sectors suggest a pattern warranting further analysis.

Client 394's transactional behaviour indicates a pattern of high mobility, evidenced by expenditures of over \$6,000 in taxis and limousines, nearly \$3,700 in tolls and bridges, and \$4,099 spent at drinking places, alongside purchases of construction materials. The potential for fund shifting under the guise of transportation or entertainment expenses is a consideration.

Client 17, seemingly less conspicuous, exhibits a broad spending pattern across essential retail categories, punctuated by unexpected refunds in specialized sectors, such as semiconductors and rail travel. This pattern, while not immediately alarming, necessitates a closer examination in the context of the other observed anomalies.

These observations highlight potential anomalies in client transactional behaviour. Further analysis is warranted to determine the underlying causes and potential risks associated with these patterns.

Implications

- ❑ Potential for Financial Irregularities: The observed transaction patterns strongly suggest the presence of financial irregularities, including potential money laundering, fund shifting, and purchase reversal schemes.
- ❑ Disguised Transaction Activities: The concentration of spending in specific categories indicates potential attempts to disguise the nature of transactions and obscure the movement of funds.
- ❑ Refund Anomalies as Indicators: The presence of unusual refunds across diverse sectors serves as a potential indicator of money cycling or deliberate attempts to manipulate transaction records.
- ❑ Need for Enhanced Monitoring: The identified anomalies call for enhanced monitoring and investigation.
- ❑ Possible Concealment of Illegal Activity: The various spending patterns could be a guise for illegal activity.
- ❑ Increased Risk of Regulatory Scrutiny: These patterns increase the risk of regulatory scrutiny for the bank.

Multi-State Late-Night Transaction Patterns

Transaction data reveals a group of clients conducting transactions across numerous states during late-night hours (midnight to 5 AM), deviating significantly from typical consumer behaviour. Client 386 exhibits the most extensive activity, transacting in 24 states. Five other clients – 480, 460, 278, 98, and 100 – follow closely, each transacting in 23 states. Clients 462 and 402 transact in 22 states, and others maintain a presence in 20 or more states.

The scale of this multi-state activity during unusual hours suggests potential involvement in activities beyond routine business or personal spending.

Implications:

- ❑ Potential for Organized Financial Crime: The extensive multi-state activity during late-night hours raises strong concerns about potential involvement in organized financial crime, such as money laundering or fraud.
- ❑ Complex Fund Movement: The widespread geographical distribution of transactions suggests a complex network of fund movement, potentially aimed at obscuring the origin and destination of funds.
- ❑ Possible Involvement in Illicit Trade: The late-night, multi-state activity could indicate involvement in the movement of high-value illicit goods or services.
- ❑ High Risk of Fraud: The sheer scale of the operation, and the time that it is taking place, indicates a very high risk of fraud.
- ❑ Need for Urgent Investigation: Aurora Bank should carry an urgent investigation into these patterns to determine the nature of these activities and mitigate potential risks.
- ❑ Possible Need for Regulatory Reporting: The scale, and the time of the activity, could require regulatory reporting.

Strategic Implications from Customer Spending Patterns

The analysis of customer spending habits, income brackets, age segments, and debt-to-income ratios provides **key strategic insights** for the bank. These insights can guide **product development, risk management, and marketing strategies** to better serve customers while ensuring financial sustainability.

1. Tailored Financial Products Based on Income Segments

- **High-income customers spend the most** on average, suggesting they may benefit from **premium banking services**, investment products, and wealth management solutions.
- **Low-income customers have lower spending power but high financial needs**, making them prime candidates for **microloans, budget-friendly savings plans, and financial literacy programs**.
- **Middle-income customers exhibit moderate spending behavior** and may be interested in **credit-building products, mid-tier credit cards, and flexible loan options**.

2. Risk-Based Credit Offerings and Debt Management Solutions

- Customers with **high debt-to-income (DTI) ratios** and **low credit scores** pose **significant default risks**. The bank can implement:
 - **Stricter lending criteria** for high-DTI customers.
 - **Debt restructuring programs** or financial coaching to help at-risk borrowers regain stability.
 - **Early warning systems** to identify customers at risk of default and intervene before they miss payments.

3. Age-Specific Financial Strategies

- **Middle-aged customers account for the highest total spending**, making them a key segment for **mortgages, retirement planning, and investment products**.
- **Seniors have lower overall spending but higher average transaction values**, indicating potential demand for **long-term care savings plans and estate planning services**.
- **Young customers (under 35) have the highest average spend per transaction**, suggesting they may be inclined toward **buy-now-pay-later (BNPL) services, credit cards, and digital banking solutions**.

4. Gender-Based Financial Solutions

- Previous insights showed **women prioritize financial security**, while men may take on **higher-risk investments**. The bank can develop:
 - **Savings and investment products tailored for women**, such as goal-based savings accounts and lower-risk investment options.
 - **Higher-risk investment portfolios and business loans for men**, catering to their tendency toward risk-taking.
 - **Customized financial advisory services based on spending behaviors and risk appetite**.

5. Enhancing Digital and Data-Driven Banking

- With **youth spending more per transaction**, digital banking should be **optimized for seamless mobile transactions, contactless payments, and AI-driven financial recommendations**.
- AI and machine learning can help identify **spending patterns and predict financial needs**, allowing for **personalized banking experiences**.

Customers with Stagnant PINs

To mitigate this vulnerability, Aurora Bank should consider implementing a mandatory PIN change policy, sending periodic reminders to customers, and analyzing transaction data to identify inactive accounts. Bolstering security measures, such as multi-factor authentication, can further contribute to reducing these risks.